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Masterarbeit

Fault-Resilient Robot Manipulation via Online Reinforcement Learning under Encoder Calibration Bias

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Abstract

Encoder calibration bias in industrial robots introduces a systematic error that simultaneously corrupts both the control loop (via incorrect Jacobian computation) and the state observation (via incorrect forward kinematics), degrading the robot from a fully observable Markov Decision Process (MDP) to a Partially Observable MDP (POMDP). This thesis addresses the challenge of maintaining robust manipulation performance under unknown encoder bias without requiring downtime for recalibration.

We first formalize the encoder bias problem as a POMDP, where the bias vector acts as an episode-level latent variable that renders the true joint state unobservable. Through observability analysis, we show that standard proprioceptive observations are insufficient to distinguish different bias conditions, and that augmenting the observation with an external end-effector position signal restores approximate observability—effectively reducing the POMDP back to an MDP that can be solved with standard reinforcement learning algorithms.

Building on this insight, we propose a two-component system: (1) a *robust task policy* trained with Reinforcement Learning from Prior Data (RLPD) under randomized encoder bias, which achieves 92–100% success rate across the full bias range $[0, 0.3]$ rad on a simulated pick-and-place task; and (2) a *backup safety policy* that protects human operators during human-in-the-loop training by performing active collision avoidance when a hand enters the shared workspace. Domain randomization on the backup policy bridges the sim-to-real gap for noisy hand detection.

Ablation studies confirm that the external position signal is necessary for bias-robust performance, validating the theoretical observability analysis. We further present a real-robot deployment architecture for the Franka Panda, including a dual-injection-point fault injection scheme that faithfully reproduces the encoder bias causal chain on physical hardware.

Contents

List of Figures

List of Tables

List of Listings

List of Algorithms

1 Introduction

1.1 Background and Motivation

1.2 Problem Statement

1.3 Contributions

1.4 Thesis Organization

2 Related Work

2.1 Fault-Tolerant Robot Control

2.2 POMDP Solution Methods

2.3 Robust Reinforcement Learning

2.4 Human-in-the-Loop Reinforcement Learning

2.5 Safe Reinforcement Learning and Collision Avoidance

3 Encoder Bias Modeling and POMDP Formulation

3.1 Encoder Calibration Bias in Industrial Robots

3.2 Causal Chain: One Fault Source, Three Effects

3.3 POMDP Formalization

3.4 Observability Analysis

3.5 Comparison with Alternative POMDP Solutions

3.6 Simulation Implementation

4 Observation-Augmented Robust Task Policy

4.1 Observation Space Design

4.2 Randomized Bias Training

4.3 RLPD Training Framework

4.4 Network Architecture

4.5 Reward Design

5 Backup Safety Policy for Human-in-the-Loop Training

5.1 Problem Definition

5.2 Scenario Design

5.3 Obstacle Motion Modes

5.4 Reward Function Design

5.5 Domain Randomization for Sim-to-Real Transfer

5.6 Sim-to-Real Alignment

6 Simulation Experiments and Analysis

6.1 Experimental Setup

6.2 Encoder Bias Degradation (H1)

6.3 Fixed-Bias Policy (H2)

6.4 Random-Bias Policy (H3)

6.5 Observation Space Ablation

6.6 Backup Policy Experiments

6.7 Integrated System Evaluation

7 Real Robot Deployment and Validation

7.1 Hardware Architecture

7.2 Fault Injection on Real Hardware

7.3 OSC vs Impedance Control Equivalence

7.4 External Localization

7.5 Experimental Results

8 Conclusion and Future Work

8.1 Summary

8.2 Limitations

8.3 Future Work

Declaration

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